Problem 5

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Problem 1 *Let f and g be two functions such that $\lim_{x\to\infty} f(x) = \lim_{x\to\infty} g(x) = \infty$.*

*We say that f **grows faster** than g as x goes to ∞ if $\lim_{x\to\infty} \frac{f(x)}{g(x)} = \infty$, or, equivalently, if*

$$\lim_{x\to\infty} \frac{g(x)}{f(x)} = 0.$$

If the limit exists, namely, if $\lim_{x\to\infty} \frac{f(x)}{g(x)} = M$, for some positive number M,

*then we say that the functions f and g have **comparable growth rates**.*

In other words, we compare the growth rates of functions f and g by computing the limit of the form $\frac{\infty}{\infty}$: $\lim_{x\to\infty} \frac{f(x)}{g(x)}$.

For each of the following pairs of functions, determine which of the pair grows faster or state that they have comparable growth rates. Justify your answer by computing the limit $\lim_{x\to\infty} \frac{f(x)}{g(x)}$.

(a) $b^x; x^x; b > 1$

(b) $x^x; \left(\frac{x}{e}\right)^x$

(c) $x^3; x^3 \cdot \ln(x)$

(d) $a^x; b^x; 0 < a < b$

(e) $\log_a(x); \log_b(x); 1 < a < b$

(f) $\ln^3(x); x^{1/2}$

(g) $x; \ln(x)\sqrt{x}$

(h) *Challenge:* $x^{40}; 1.004^x$ *(Hint: Use the substitution $x = \ln(t)$.)*

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(i) *Put the following functions in order of growth rate.*

$$x^3 \cdot \ln(x), \ln^3(x), x^x, 1.004^x, x^{40}, x^3$$